

Sneha bejugam

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of experience designing and deploying scalable **Machine Learning** and **Generative AI** solutions across fintech and healthcare domains. Experienced in building **LLM**, **RAG**, and predictive ML applications using **Python**, **LangChain**, **LlamaIndex**, **LightGBM**, **Azure Machine Learning**, and **AWS Bedrock**. Skilled in end-to-end model development, feature engineering, MLOps, cloud deployment, and data engineering using **Azure Databricks**, **Apache Spark**, and **FastAPI**. Proven ability to deliver production-grade AI solutions that improve business outcomes through automation, intelligent search, and predictive analytics.

TECHNICAL SKILLS

- Programming:** Python, SQL
- Generative AI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI, AI Assistants, Function Calling
- LLM Frameworks:** LangChain, LangGraph, LlamaIndex
- Machine Learning:** Scikit-learn, LightGBM, Feature Engineering, Model Training, Hyperparameter Tuning (Optuna), Model Explainability (SHAP), Classification, Predictive Modeling
- Cloud & MLOps:** AWS Bedrock, Azure Machine Learning, MLflow, Docker, Kubernetes, Amazon EKS, CI/CD, Git, GitHub Actions
- Data Engineering:** Azure Data Factory, Azure Databricks, Apache Spark, ETL Pipelines, Data Preprocessing, Pandas, NumPy
- Vector Search & Databases:** Pinecone, Amazon OpenSearch, PostgreSQL, Semantic Search, Embedding Models, Vector Search
- Backend & APIs:** FastAPI, REST APIs, AWS Lambda, Amazon S3, API Integration
- Monitoring & Visualization:** LangSmith, Amazon CloudWatch, Power BI, Model Monitoring, Performance Evaluation

PROFESSIONAL EXPERIENCE

MasterCard, NY | AI Engineer **Jan 2025 – Present**

Project: *Generative AI Merchant Support Assistant*

- Developed an enterprise **Generative AI Merchant Support Assistant** using **Python**, **LangChain**, and **GPT-4** to automate responses for merchant queries related to payment processing, chargebacks, settlements, and API integrations.
- Built a **Retrieval-Augmented Generation (RAG)** pipeline with **LlamaIndex** and **Pinecone**, enabling accurate retrieval of information from merchant documentation and reducing irrelevant responses by **35%**.
- Designed semantic search workflows using **Amazon OpenSearch** and embedding models to improve document retrieval accuracy and deliver context-aware responses with low latency.
- Developed scalable backend APIs using **FastAPI** to integrate the AI assistant with internal merchant support applications, supporting secure and real-time query processing.
- Deployed containerized AI services on **AWS Bedrock**, **Docker**, and **Amazon EKS**, improving application scalability, reliability, and deployment efficiency across environments.
- Automated document ingestion and preprocessing pipelines using **AWS Lambda** and **Amazon S3**, enabling continuous indexing of payment policies, FAQs, and technical documentation.
- Implemented **MLflow** for prompt versioning, experiment tracking, and response evaluation, improving model performance consistency and reducing testing effort by **30%**.
- Monitored application performance using **LangSmith** and **Amazon CloudWatch**, analyzing LLM responses, optimizing prompt quality, and maintaining **99.9%** service availability.

Accenture, India | ML Engineer

Aug 2020 – Nov 2023

Project: *Enterprise Medication Adherence Prediction Platform*

- Engineered scalable data pipelines using **Azure Data Factory** to integrate pharmacy claims, EHR, prescription, and member datasets, automating ingestion of over **20M+ healthcare records** and improving data availability for downstream ML workflows.
- Built advanced predictive models using **LightGBM** to identify patients at high risk of medication non-adherence, improving prediction performance by **18%** over the baseline model through feature optimization and class imbalance handling.
- Developed robust feature engineering workflows using **Pandas** and **NumPy** to create adherence history, medication refill behavior, chronic disease, and demographic features, increasing model accuracy and reducing feature redundancy.
- Optimized model performance using **Optuna** for automated hyperparameter tuning, reducing training iterations by **35%** while improving validation stability across multiple patient cohorts.
- Performed model explainability and clinical validation using **SHAP**, enabling healthcare teams to interpret prediction drivers and support evidence-based patient outreach strategies with greater transparency.
- Built distributed data preprocessing and model training pipelines using **Apache Spark** on **Azure Databricks**, reducing large-scale data processing time by **45%** for millions of patient records.
- Managed model training, versioning, and deployment workflows using **Azure Machine Learning**, enabling reproducible experimentation and accelerating model release cycles across development and testing environments.
- Designed interactive adherence risk dashboards using **Power BI**, providing care management teams with actionable patient risk insights and key model performance metrics, helping prioritize interventions that contributed to an estimated **12% increase in medication adherence** among targeted patient populations.

PROJECTS

FashON: AI-Powered Personal Styling App | (Flutter, MVVM, Firebase, OpenAI API)

- Built an AI-powered **Flutter app** for iOS and Android, enabling virtual wardrobe management, outfit recommendations, and outfit compatibility analysis using ML models, with outfit visualizations generated via the **OpenAI DALL-E 2 API**.
- Architected the app using MVVM and implemented secure authentication and access control with Firebase Authentication and Firestore security rules for user-level data isolation.

FabricFlow: Generative AI Fashion Content Platform | (Python, Gemini 2.0 Flash, ElevenLabs, FastAPI, AWS)

- Built a multimodal AI platform that transforms fabric images into photorealistic model visuals, bilingual captions, and Instagram-ready reels using **Gemini 2.0 Flash** and **ElevenLabs**, reducing content creation time from 2 hours to 30 seconds.
- Developed an AI orchestration pipeline on **AWS** for real-time content generation, integrating image processing, Hindi voiceovers, and automated social media content delivery.

EDUCATION

M.S in Computer Science – Syracuse University, Syracuse, New York

Jan 2024 – Dec 2025

B.Tech. in Computer Science and Engineering – KMIT, Hyderabad, India

Jul 2018 – Jun 2022

CERTIFICATIONS

- AWS Certified Developer Associate** - (AWS Academy, 2025)
- Snowflake** - (Snowflake Academy, 2023)
- (hands on essentials - **data engineering**, hands on essentials - **data lake**, hands on essentials - **data warehouse**)